Tab 2

Signal or Noise - The Gateway Stats Data Sprint

Cover & Metadata

FIELD	SUMMARY
Course / topic	Data-Informed Teaching / Academic Data Literacy for Lectur analysis-procedure selection on sample data
Simulation title	Signal or Noise: The Gateway Stats Data Sprint
Designer / Builders	[Name — to be completed by production team]
Date / version	[Date, version 1.0]
Target learners	University lecturers and faculty (early- to mid-career); assur knowledge limited to basic descriptive statistics (mean, corr teaching or research-methods exposure — no advanced sta
Estimated duration	35–45 minutes, self-paced. No section is hard-timed; Scene left" framing is narrative pressure, not a real-time countdow
Setting	Northbridge University (fictional) — the Gateway Statistics 1 run across six parallel sections

Simulation Type Selector

☑ Dialogue / branching

☑ Evidence-gathering / environment-emergent exploration

☐ Instrumented testing

☑ Procedural sequencing

☑ Token / resource allocation

☑ Drag-and-drop / annotation builder

☐ Others _____

Tone, voice & reading level

Professional, collegial, direct — a peer speaking to peers, not an instructor speaking to students. Plain English; every statistical term (correlation, confound, covariate, regression) is defined in one plain sentence the first time it appears, then used without re-explanation. Second-person address ("you"). No filler AI-style phrasing, no rhetorical hand-holding. International/neutral English spelling.

Introduction

Signal or Noise: The Gateway Stats Data Sprint is an interactive simulation that places you in the role of a Data-Informed Teaching Fellow at Northbridge University. When pass rates across six Gateway Statistics 101 sections suddenly fall, you are given grades, attendance, LMS engagement data, and student feedback to investigate what happened.

Across five decision points, you will decide which analysis to use, what the evidence can actually support, and how confidently the findings should be communicated. Your choices will shape the evidence available to you, your relationship with key stakeholders, and the recommendation you ultimately present to the Programme Review Board.

The objective is not to find the fastest answer. It is to distinguish signal from noise, match each question to an appropriate analytical procedure, recognise confounding and correlation, and make a recommendation that remains defensible when the evidence is scrutinised.

The Premise

The setting / world	Northbridge University's Gateway Statistics 101 — a required first-year course d six parallel sections with a shared syllabus and a common final exam, taught by tenured faculty and adjuncts.
The learner's role	You are the newly appointed **Data-Informed Teaching Fellow** for the Gateway P role created to help departments use their own data well. You report jointly to th Dean and the Faculty Senate's Teaching & Learning Committee.
The problem / situation	Pass rates across the six sections have fallen from 78% to 61% in a single seme Associate Dean has pulled four datasets — section grades, attendance logs, LM engagement, and end-of-module survey comments — and wants answers befo Programme Review Board, where funding for two adjunct lines is on the agenda
The goal / objective	Select and apply sound analysis procedures — not just any procedure — to iden actually driving the decline, and deliver a recommendation the Board can act on misreading the data.

State & Scoring Framework

ACCURACY / STATE STRANDS	
Strand 1 — Analytical Rigour	Measures whether the procedure chosen at each decision p data type and the question being asked. Updates in Scenes
Strand 2 — Causal Caution (Statistical Integrity)	Measures whether the learner distinguishes correlation fron avoids premature causal claims. Updates in Scenes 2, 3, and
Strand 3 — Stakeholder Trust	Measures how the learner's pace, diplomacy, and evidence-h the Dean, the Faculty Senate representative, and the Assess Updates in Scenes 1, 4, and 5.
Running log	**Analysis Log** — auto-populates with each procedure the lea a one-line rationale. It is visible throughout but not separate however, its entries are the only material available to build th so weak earlier choices constrain what can be argued convi end.

Structure & Flow

Standard MIVA Shell: MIVA University Opening → Optional Intro Sequence → Title / Landing Page / Student Name → Simulation Beats 1–5. The opening shell does not count toward the five primary simulation beats.

Scene 1 — The Alarm

NODE 1 — Six Sections, One Number	
Situation	You open the shared dashboard the Associate Dean has sent. It shows one stark lin → 61%, same course, same final exam, six sections. Attached are four datasets you section-level grades, attendance logs, LMS engagement/clickstream, and end-of-m comments. The Dean's cover note just says, "Where do we start?"
Interaction	*Drag-and-drop ranking.* Four candidate "first moves" appear as cards: (a) Compare adjunct-taught sections directly against the four faculty-taught sections; (b) Run statistics on grade distributions across all six sections before comparing anything; attendance records with final grades immediately; (d) Interview students in the two lowest-performing sections first. Learner drags the cards into 1st–4th priority order
Response / indicator	Ranking (b) first unlocks a clean set of distribution charts and logs "Baseline establ (a) or (c) first instead triggers a flag: the dashboard highlights that section sizes ar baseline exists yet, and the Dean's note updates to "Let's not jump to blaming anyo (a) first also costs Stakeholder Trust, since it reads as targeting adjuncts before ev
Teaching content	The difference between exploratory/descriptive analysis (establishing what the dat inferential or comparative analysis (testing whether a difference is real). Skipping st comparison or a correlation before establishing a baseline is a common, costly orde manufacture a story the data doesn't yet support. Delivered as an unlock conseque lecture, so the cost of skipping ahead is felt, not just told.
State change	Analytical Rigour +2 if (b) ranked first; +0 otherwise. Stakeholder Trust −1 if (a) rank Log records the chosen first move and rationale.
Goes to	→ Node 2

Scene 2 — The Correlation Trap

NODE 2 — Two Numbers, No Story Yet	
Situation	The baseline is in. Two correlations jump out: attendance vs. final grade (r = 0.42), and LI video-watch-time vs. final grade (r = 0.51). Meanwhile, survey comments repeatedly men assignment instructions." A Faculty Senate representative messages you: "So it's attend we just mandate it?"
Interaction	*Sort into bins.* Six candidate next-step cards — including "Run a multiple regression cont GPA and section," "Declare attendance the cause and recommend a mandatory-attenda a one-way ANOVA comparing the six sections' pass rates," "Interview students in the two "Report the two correlations to the Board as the explanation," and "Check whether prior across sections before comparing them" — are sorted by the learner into "Sound Next St / Flawed."
Response / indicator	Correctly binning "declare attendance the cause…" and "report the two correlations as th as Premature/Flawed reveals a hidden data point: prior-semester GPA is unevenly distrib six sections (sections were not randomly assigned). This confound only surfaces once t flagged the causal-leap cards as flawed — reasoning is rewarded with new evidence, not
Teaching content	A correlation coefficient describes association, not cause. When groups being compared variable that also predicts the outcome (here, prior GPA), that variable is a *confound* — a controlled for, typically via regression, before attributing an effect to attendance or enga
State change	Causal Caution +2 for each correctly binned "flawed" causal-leap card; −1 for each misca Analytical Rigour +1 if "check prior GPA across sections" and "run multiple regression" are Sound. Analysis Log records the confound.
Goes to	→ Node 3

Scene 3 — Controlling for Confounders

NODE 3 — Isolating What Actually Matters	
Situation	You now know prior GPA differs by section, so raw comparisons are unsafe. The Assessm "Which factor most explains the drop, once you account for that?" Four possible analysis table, each a short labelled path.
Interaction	*Path drawing / route-the-line.* The learner draws a line from the research question to one Path A — t-test comparing two sections directly; Path B — chi-square test of pass/fail by — one-way ANOVA across six sections, no covariates; Path D — multiple linear regression as outcome and prior GPA, section, and survey-derived "instruction clarity" score as pred path can be confirmed.
Response / indicator	Drawing Path D unlocks a coefficient table: once prior GPA and section are controlled for, clarity" is a significant predictor of final grade, while attendance's apparent effect shrinks Paths A–C connect but return an on-screen note that the chosen test cannot hold a con covariate like prior GPA constant, so the result is flagged "confounded — not yet usable," is returned to choose again.
Teaching content	Matching procedure to question and data type: t-tests and chi-square compare two mea categorical counts but can't hold a continuous variable constant; ANOVA compares more means but, without covariates, still can't separate GPA's influence from section's. Multipl hold prior GPA and section constant while estimating the independent contribution of in
State change	Analytical Rigour +3 if Path D is ultimately chosen. Analysis Log records the key finding: " clarity, not attendance, is the stronger controlled predictor."
Goes to	→ Node 4

Scene 4 — The Pressure Point

NODE 4 — One Token, Two Days Left	
Situation	Two days remain before the Programme Review Board. You have budget and time for exa piece of analysis. The Dean wants a fast, defensible answer. The Faculty Senate rep wor be blamed. The Assessment Office quietly suggests checking whether the final exam its term.
Interaction	*Resource allocation (token drag).* One "analysis token" is dragged into exactly one of fou "Qualitative coding of open-ended survey comments," "Follow-up focus group with two s "Item-level analysis of the final exam itself," or "Re-run the regression on three additiona semesters for robustness."
Response / indicator	Allocating the token to item-level exam analysis reveals that this term's exam added a ba pilot data show are ambiguously worded — a confound nobody had considered, and one explains part of the drop independent of instruction or attendance. The focus-group opt but slow, low-generalizability data with the two days now nearly gone. The historical-reg "safe" — it confirms Scene 3's finding is stable — but does not test the live exam-validity Assessment Office flagged.
Teaching content	Choosing an analysis procedure is not only a statistical decision — it is also a resourcing under real time and cost constraints, and it requires weighing which remaining unknown decision-relevant, not which is most interesting or most reassuring.
State change	Analytical Rigour +2 and Causal Caution +2 if exam item-analysis is chosen (it directly te unexamined confound). Stakeholder Trust +1 if the Analysis Log shows a stated trade-of whichever option is chosen; −1 if it does not. Analysis Log records the choice and what re going into Scene 5.
Goes to	→ Node 5

Scene 5 — The Recommendation

NODE 5 — What the Board Hears	
Situation	You're in the Programme Review Board meeting. Your Analysis Log — built from every cho open beside you. You must label each finding correctly and attach it to a recommendatio present, or risk the Board overreacting to a claim your data doesn't support.
Interaction	*Annotate and label.* Four finding chips, drawn from the learner's own Analysis Log ("atten correlation," "instruction-clarity effect (controlled)," "exam item ambiguity," "prior-GPA i sections") are each dragged onto an epistemic label ("Evidence-based / actionable," "Co needs further study," "Confirmed confound — must be controlled for") and then connect recommended action chip ("Revise ambiguous exam items," "Clarify assignment instruct mandate attendance yet — commission a controlled follow-up," "Adjust the section-assi process").
Response / indicator	Correct labelling assembles a calibrated recommendation package the Board adopts larg presented. Mislabelling any finding as "Evidence-based / actionable" when it is only corre commonly, dragging "attendance correlation" straight to an action chip — triggers the br Dean announces a mandatory-attendance policy on the spot: a visible, on-record overre revisits, and one that cannot be undone within the simulation.
Teaching content	Communicating findings to non-technical stakeholders responsibly means keeping the e of each claim visible in the recommendation itself, not just in the analysis behind it — "thi correlational" has to survive translation into the room, or the room will act as if it never he
State change	Causal Caution +3 and Stakeholder Trust +2 for a fully correct label/action match. Each m Causal Caution −2 and can trigger the Overreach flag regardless of the numeric score els Analysis Log locks; final strand tallies are calculated.
Goes to	→ Debrief

Audio/Video Production Spec

Format & duration	Self-paced interactive module, ~35–45 minutes. Primarily UI-driven (dashboards, ch messages, drag interactions) with light narrated voiceover for key story beats — the dashboard note, the Faculty Senate message, and each scene's opening Situation t
Voices required	Narrator — neutral, calm, brisk pace, guides transitions. Associate Dean — measured not unkind. Faculty Senate Representative — concerned, protective of adjunct colle hostile. Assessment Office — dry, precise, minimal affect.
Visual style	Flat, UI-illustrated style — dashboards, charts, and chat-style messages rather than heavy animation. Chosen because the "evidence" (distribution charts, coefficient ta stay crisp and legible under learner scrutiny, and because a UI aesthetic matches th premise that the learner is working inside real institutional tools.
Accessibility	Default-on subtitles/captions for all voiced content, minimum 18px, high contrast.
Fixed vs. open	**Fixed:** the five decision points, their correct/flawed branches, and the strand-scorin change. **Open:** dashboard visual styling, character portraits, exact line-reads, and background/ambient sound design are producer's creative latitude.

Full transcript (sample — Opening & Scene 1)

[AUDIO CUE: soft notification chime]

NARRATOR: "You've just started as Northbridge University's Data-Informed Teaching Fellow. Your first message is already waiting."

[AUDIO CUE: incoming-message sound]

ASSOCIATE DEAN (VO, measured, slightly urgent): "Six sections, same exam, same syllabus — and pass rates dropped from seventy-eight percent to sixty-one in one semester. I've pulled four datasets for you: grades, attendance, LMS engagement, and the end-of-module survey. Where do we start? I need something for the Board by Friday."

[AUDIO CUE: dashboard UI sound — panel opens]

NARRATOR: "Four possible first moves are on the table. Only one of them gives you a baseline before you start comparing anything. Rank them in the order you'd actually run them."

[Remaining scenes: the word-for-word narration for Scenes 2–5 is the "Situation" text in each Section 6 node table above, formatted for voiceover in the same register as this sample. Production should expand each Situation field into full VO copy following this pattern rather than duplicating it here.]

Debrief & Leaderboard

How tracked strands combine into a final score: Each strand (Analytical Rigour, Causal Caution, Stakeholder Trust) is normalized to a 0–100 scale, then combined as a weighted composite — but a single flag from Scene 5 can override the numeric result, so scoring is deliberately not simple or purely additive.

Combined score formula:

Final Score = (Analytical Rigour × 0.40) + (Causal Caution × 0.35) + (Stakeholder Trust × 0.25)

Worked example:

Analytical Rigour = 85, Causal Caution = 60, Stakeholder Trust = 70

Final Score = (85 × 0.40) + (60 × 0.35) + (70 × 0.25) = 34 + 21 + 17.5 = **72.5**

Branching outcome matrix (final ending depends on strand *bands*, not the numeric score alone):

Rigor	Causal Caution	Trust	Ending
High	High	High	**"The Trusted Data Lead"** — the Boar recommendations in full; you're invite Fellow role.
High	High	Low	**"Right Answer, Wrong Room"** — you correct but delivered so bluntly or lat tables the recommendation for a mor
Any	Overreach flag triggered in Scene 5	Any	**"Public Misstep"** — a policy (e.g., mar attendance) is announced based on didn't support; you correct the record damage is on the minutes.
Low	Any	Any	**"Popular but Wrong" / "Back to the** the recommendation feels reasonabl mismatched procedure; the Assessm

Rigor	Causal Caution	Trust	Ending
			re-runs the analysis after the meetin

What's displayed but not scored: the Analysis Log itself (visible throughout, and the raw material for Scene 5, but not separately graded); character dialogue flavour text and reactions.

Backend (Supabase/Firebase), fields tracked: user_id, per-scene strand deltas (rigor, caution, trust), overreach_flag (boolean), final_score, ending_id, time_spent, analysis_log (JSON array of chosen procedures + rationale), completion_timestamp.

Facilitator dashboard needs: cohort-level distribution of ending_id; the most-missed decision point (e.g., % of learners who correctly chose Path D in Node 3); average strand scores by cohort; ability to filter by department or cohort intake.

Accessibility Requirements

☐ Default-on subtitles for all voiced content, minimum 18px, high contrast

☐ Keyboard or tap/click alternative for every drag/hold gesture

☐ Text-label equivalents for directional/compass navigation

☐ Accessible labelling for timers and data-entry fields

☐ Minimum 44px touch targets throughout

☐ Colour-independent bin/label distinctions — icons or patterns, not colour alone, differentiate "Sound" vs. "Premature" bins and epistemic labels in Scenes 2 and 5

☐ Click-to-select-then-click-to-place alternative for all path-drawing (Scene 3) and token-allocation (Scene 4) interactions, in addition to drag

Tab 1

DATASET ANALYSIS SANDBOX

Behind the Order

A Dataset Analysis Sandbox Simulation for Public Health Policy, Law & Ethics

Colour Legend for Developers

Colour		Meaning	Used for
Deep Blue		Act I — Individual Case	Scene headers, decision cor tables
Gold		The Bridge	Transition screen between a
Green		Act II — Cohort / Population	Scene headers, decision cor tables
Purple		Bonus / Optional	Unlockable branching scene
Red		Feedback / Risk callouts	Every scene's feedback box, warnings
Slate Grey		Facilitator / developer notes	Build notes, scoring matrix h

Introduction

The Compliance Dataset Sandbox is a fictional public-health legal-compliance environment used throughout this simulation. It holds two views of the same underlying world: a single-case view, where one patient's isolation order is reviewed in isolation, and a cohort view, where that same order is seen alongside hundreds of others across a county. The simulation's entire argument is that these two views ask different questions, draw on different bodies of law and ethics, and — critically — that the first view's documentation becomes the second view's evidence, whether or not the analyst realises it at the time.

The domain is a county health department's predictive compliance tool, ComplianceWatch, which flags patients under communicable-disease control orders (isolation, quarantine, directly observed therapy) as "high risk" of non-compliance, prioritising them for escalated legal enforcement, up to and including a court petition for involuntary detention. This is a deliberate choice of setting: public health law sits exactly at the fault line between the state's police power to protect the community and an individual's constitutional liberty interest — and a predictive score that quietly tilts that balance is invisible at the level of one case and unmistakable at the level of a population. The lessons transfer directly to any public-health legal context where an algorithm hands an analyst a risk percentage and asks them to decide how much liberty to restrict.

Learning Objectives

- Apply the least-restrictive-means principle and the Jacobson-style public-health police-power balancing test to a real, evidence-thin case — not as an abstract doctrine, but as a live documentation decision.
- Distinguish a predictive compliance score from a legal and ethical judgement, and identify when a score is being used as a shortcut around due process rather than an input to it.
- Triangulate legal, clinical, and social evidence (structural access barriers, language, transport, prior compliance history) before recommending an enforcement pathway.
- Navigate a cohort-level legal-compliance dataset to detect disparate impact invisible in single-case review, and distinguish when a question needs individual evidence versus population evidence.
- Recognise the specific mechanism by which one analyst's case documentation becomes evidence in a health-equity legal complaint three months later — and where in their own workflow that link currently breaks.
- Build an enforcement decision rule (individual or systemic) and pressure-test it against contradicting legal, ethical, and equity evidence.
- Communicate a legal/ethical judgement at the register appropriate to who receives it: a patient, a community advocate, or a health-equity oversight board.

- Anticipate second-order effects of a compliance decision: on one patient's liberty and trust in public health, on a formal disparate-impact complaint, on institutional legal precedent, and on the analyst's own defensibility if challenged at either scale.

Scenario Overview

Act I — First Person Singular. You are a Public Health Law & Ethics Analyst in a county Compliance Review Unit. Three days ago, ComplianceWatch flagged Mr. Tunde Balogun — a patient under an active tuberculosis isolation order — at 88% risk of non-compliance. You have five working days before a compliance hearing to recommend enhanced directly-observed therapy (DOT) or a court petition for involuntary detention. What you document now, not just what you decide, becomes a permanent legal record.

The Bridge — Three Months Later. A short interstitial shows your Act I case file being pulled, unopened, into a much larger legal-compliance dataset. You do not control what happens next; you only find out how well your documentation held up.

Act II — First Person Plural. You have been given analyst access to the county-wide Compliance Dataset Sandbox. A civil-liberties and health-equity coalition has filed a formal disparate-impact complaint on behalf of eleven patients — Mr. Balogun among them, all recent immigrants and/or non-English-speaking — alleging ComplianceWatch systemically pushes this group toward detention petitions rather than less restrictive alternatives. A county Health Equity & Legal Oversight Board meets in 72 hours. Your Act I case file is now Exhibit 7. Whether it helps or hurts the eleven patients' complaint depends on choices you already made, and cannot unmake.

Learner Role(s)

Phase	Role
Act I	Public Health Law & Ethics Analyst — first-line reviewer of an algorithmic compliance flag, pers for the legal case file that will outlive the hearing.
The Bridge	No active role — observer of your own file being reused. A deliberate design choice.
Act II	Analyst, county-wide Compliance Dataset Sandbox. May be the same person three months o choice — a different colleague, changing the emotional register of finding your own prior work

Framework Mapping (Developer Reference)

Every scene lives inside the single Dataset Analysis Sandbox environment. The interaction 'modules' below are how the required ISP framework list is expressed inside that environment — never as isolated mini-games, always as an operational task within the same case-file / dataset interface.

Scene	Sandbox Module (what the learner sees)	ISP Framework
1. The Flag Arrives	Triage queue — drag patients into legal-action tiers	Resource Allocation / Prioritisati
2. Reading the Score	Score breakdown panel — sort & annotate features	Annotate and Label
3. The Paper Trail	Case timeline — link records, attach or summarise	Connect the Dots / Timeline Placement
4. The Home Visit	Interpreter-assisted dialogue — choose opening, record framing	Dialogue Choice / Voice-Speech Input
5. Weighing It All	Four-value balance dial	Tilt and Balance
6. The Decision	Rule-builder — evidence → recommended pathway	Conditional Simulator
Bridge	Read-only case file reveal	Passive Reveal (no framework — design)
7. First Login	Cohort dashboard — sort six summary statistics	Sort into Bins / Categorise
8. Filtering the Cohort	County map — pin flagged cases against transit & clinic access, then predict the pattern	Map Pin Drop + Sketch and Recognise
9. The Advocate Calls	Phone-call dialogue tree	Dialogue Choice / Role-Play
10. Competing Deadlines	72-hour resource dial across four work-streams	Resource Allocation
11. Building the Rule	Policy-jigsaw — assemble legal threshold, safeguard, comms plan	Stack and Build (Jigsaw) + Conditional Simulator + Rotate t Align
12. The Oversight Board	Four-week consequence timeline	Timeline Placement

Scene	Sandbox Module (what the learner sees)	ISP Framework
Bonus A – Vendor Rebuttal	Evidence-rebuttal panel	Annotate and Label / Equation-style reasoning
Bonus B – Second County	Comparative dataset	Pinch and Zoom (compare two datasets side by side)
Bonus C – Board Paragraph	Report drafting — spoken or written	Voice / Speech Input

The environment itself — the persistent, evolving case-and-cohort dataset that carries every decision from Scene 1 to Scene 12 — is the Simulation Sandbox framework. It is not one scene; it is the container all other scenes render inside.

ACT I — THE SINGLE CASE

You open the Compliance Dataset Sandbox in single-case mode. One file. One patient. Five days.

SCENE 1 — The Flag Arrives

Interaction Type	Timed Prioritisation / Resource Allocation
ISP Framework	*Resource Allocation (Problem-solving)*

The ComplianceWatch dashboard shows 10 flagged patients this week, ranked only by risk score. Mr. Balogun's is 3rd highest. Limited review time remains before the hearing calendar locks.

Task

- Drag the 10 patients into three legal-action tiers: Prepare detention petition / Enhanced monitoring this week / Routine follow-up — using only visible metadata: risk score, missed-DOT count, prior order history, primary language on file.
- Mr. Balogun: 88% risk, one missed DOT visit (day fell on a shift he could not leave), first order ever issued, primary language Yoruba, interpreter requested twice, not yet fulfilled.
- Contrast case: 91% risk, third missed visit this order, prior non-compliance finding, already under a modified order, English-speaking.

Decision Consequence

- Score-only sorting → Mr. Balogun bundled with repeat-non-compliance cases and given the same fast-tracked petition-prep review. Immediate: faster queue clearance. Delayed: every later reading of his file is coloured by the tier he was sorted into — and the file opened under this tier defaults to a two-line closure note, which is the version Act II will find.
- Context-weighted sorting → Mr. Balogun prioritised for a closer, not harsher, look, because a first-time flag with an unmet interpreter request and one work-related missed visit is a due-process gap, not necessarily a non-compliance pattern. Immediate: slightly slower clearance. Delayed: a legally defensible read — and a file with enough header detail that Act II can actually use it.
-

New Sub-Choice — Documentation Mode

Before Scene 2, choose how to log the triage decision: “Sorted per protocol, no further note” — clears the queue exhibit value later; or “Sorted to [tier] because [specific access barrier]; flagging for evidence-based review” — seconds now, is the single highest-leverage action in the entire simulation for Act II's Legal Rigor score.

Feedback

A risk score is one input to a legal threshold, not the threshold itself. Documentation habits formed under this k pressure are invisible in the moment and decisive at a hearing — or a board review — later.

Facilitator Note

Anchoring bias before evidence appears, plus: this is the first of several moments where 'what I decided' and 'w silently diverge — and the simulation is built to make that divergence cost something concrete later, rather than

SCENE 2 — Reading the Compliance Score

Interaction Type	Sort into Bins / Annotate and Label
ISP Framework	*Annotate and Label (Problem-solving)*

Six data points, presented with false equivalence:

- Overall compliance-risk score: 88%
- Missed-DOT feature: flagged “high concern” (one missed visit, reason on file: work shift)
- Interpreter-request feature: two open requests, unresolved 11 days
- Two features flagged as “pattern matches” for historical non-compliance cohorts
- Note: score generated before the interpreter request was fulfilled, so the intake interview behind the score was conducted in English
- Small-print model card disclaimer: model accuracy is degraded for patients whose intake interview was not conducted in their primary language

Task

- Sort into Strong evidence of non-compliance risk / Weak-ambiguous evidence / Not evidence at all.

Decision Consequence

- The missed-DOT feature belongs in weak/ambiguous given the documented work-shift conflict; the 'pattern match' features and the disclaimer belong in 'not

evidence' until an interpreter-conducted interview confirms or disconfirms the underlying facts.

New Sub-Interaction — Annotate for the File

For each item sorted into Strong or Weak/ambiguous, optionally attach a one-line legal rationale that travels wit
Annotated items are retrievable in Act II as individual-case evidence, not just an aggregate score. Un-annotated
the bare "88%" the cohort dataset will show — accurate, but stripped of the due-process nuance that could cle
case on reappraisal.

Feedback

The annotation habit costs almost nothing here and is worth a great deal later — this is the scene where 'the su
legally' stops being abstract and becomes a concrete artifact you either did or didn't create.

Facilitator Note

The technical heart of Act I. This scene's optional annotation is the biggest lever the simulation has: it converts
lesson ('read past the headline number') into an inter-act one ('what you wrote down about the headline numb

SCENE 3 — The Paper Trail

Interaction Type	Connect the Dots / Timeline Placement
ISP Framework	*Connect the Dots + Timeline Placement (Comprehension–Problem-solving)*

Four artifacts on a timeline:

- Order issued (day 1): standard isolation order, standard DOT schedule set without confirming shift pattern
- Day 6 clinic note: patient asked about switching DOT time, request not actioned
- Missed visit (day 11): same day as a confirmed work shift on his employer's schedule
- Social-worker narrative note (2 days pre-flag): patient "anxious about losing job if he keeps missing shifts, asked twice for evening DOT slot"

Task

- Connect artifacts showing a structural scheduling barrier versus a genuine pattern of avoidance.

Decision Consequence

- Connecting the DOT-time request to the missed visit supports 'structural-barrier-driven flag,' a materially different and lower-urgency category than

'deliberate non-compliance.' Under the least-restrictive-means principle, this connection changes which legal remedy is even available to recommend.

New Sub-Choice — Preserve or Summarise

Once connected, choose to (a) attach the full social-worker narrative note to the case file, or (b) write a summa
work conflict"). Attaching costs a small records-processing step now; only the attached version survives verbat
evidence-comparison scene. The summary survives as a claim someone else has to take on trust.

Feedback

A summary is a memory of evidence. An attachment is evidence. Under hearing-calendar pressure these feel int
they are not, and the gap between them is exactly where cases get re-opened three months later in front of an

Facilitator Note

Process evidence changes the legal category of the case, not just its tone. This preserve/summarise mechanic
Scene A in Act II.

SCENE 4 — The Home Visit

Interaction Type	Dialogue Choice / Role-Play with Voice-Speech Input
ISP Framework	*Dialogue Choice + Voice/Speech Input (Creation)*

An interpreter is now available. Three ways to open the visit — record (type or speak) your first question:

- A — Lead with the order: "The system flagged you as high-risk for missing your isolation requirements. Can you explain the missed visit?"
- B — Lead with process: "Walk me through your week — your shifts, your appointments, what's been hard to manage."
- C — Lead with undercutting reassurance: "This is probably just a mix-up, but is something making this difficult for you?"

Decision Consequence

- Under B, Mr. Balogun volunteers, through the interpreter and unprompted, that he requested an evening DOT slot twice and was never told whether it was approved — surfacing the real explanation without being led toward it. B yields the richest, least-leading evidence; A is defensible but primes the patient toward defending against the flag; C supplies the explanation before the patient can offer it, which is weakest evidentially and also the most legally fragile if challenged as leading.

New Sub-Choice — What Gets Quoted

After the visit, choose how to record it: a verbatim-adjacent paraphrase ("Patient reports two unresolved eveni primary language Yoruba, interpreter present throughout") — preserves specific, defensible reasoning; or a com ("Patient explained absence, no issue") — technically true, flattens a due-process failure into something that re context, like a dismissal.

Feedback

A label written under time pressure often outlives the interview that produced it by months — and whoever read a hearing officer or an equity board, has no way to recover the nuance you had in the room.

Facilitator Note

Investigative stance shapes evidence quality; this applies to write-ups as much as to openings. A fair interview, becomes an unfair record.

SCENE 5 — Weighing It All

Interaction Type	Tilt and Balance
ISP Framework	*Tilt and Balance (Creation)*

Four values, weighted out of 10 total: Public Health Safety, Least Restrictive Means (liberty interest), Consistency, Administrative Efficiency/Speed. Extreme allocations trigger visible consequence previews.

New Layer — Consistency Now Has Teeth

The sandbox states explicitly: "Nine other patients with similar structural-barrier profiles are currently flagged a Your weighting here will be visible to whoever reviews all nine together." This previews Act II directly, without req have played it yet.

Feedback

No clean answer; reasoning transparency is what's scored. This scene is the hinge between the two acts — wha here is what nine strangers will quietly be compared against later.

SCENE 6 — The Decision & Its Aftermath

Interaction Type	Conditional Simulator (If-Then Decision Tree)

ISP Framework	*Conditional Simulator (Problem-solving)*

Build a recommendation from evidence-category conditions: petition court for involuntary detention / enhanced DOT with an evening slot and confirmed interpreter access / close the flag with documented rationale. Then see a six-week epilogue.

New Epilogue Branch — The File Gets Pulled

Regardless of path, a final card always appears: “Three months from now, your case file for Mr. Tunde Balogun w county-wide legal review. What you documented — not just what you decided — is what that review will see.” Th score; it is a bridge, not a judgement.

Facilitator Note

Delayed-consequence principle. This closing card converts Act I from a self-contained lesson into the first half Keep it neutral, like a filing notice — because that is what it is.

THE BRIDGE — Three Months Later

Interaction Type
Passive Reveal / Read-Only Case File — no decisions here, deliberately.

The learner sees their own Act I case file exactly as it exists on the system, assembled entirely from earlier choices:

- The Scene 1 documentation note (or its absence)
- The Scene 2 annotations (or the bare 88% with nothing attached)
- The Scene 3 social-worker note (attached in full, or reduced to a summary line)
- The Scene 4 interview record (nuanced paraphrase, or flattened label)
- The Scene 5 weighting rationale (documented, or not)
- The Scene 6 final recommendation

Beneath it: "This file has been added to Exhibit Set: County Compliance Review, Case #7 of 11."

Why No Interaction
Facilitator testing found that letting learners 'fix' their Act I record retroactively defeats the entire teaching poir re-annotate the past once someone else, and their liberty, is depending on it.

Facilitator Note
Natural pause point for group sessions. Comparing Bridge screens across a room is usually the most energising workshop.

ACT II — THE COHORT

You log in to the county-wide Compliance Dataset Sandbox. Eleven patients. One complaint. Seventy-two hours.

SCENE 7 — First Login

Interaction Type	Sort into Bins
ISP Framework	*Sort into Bins / Categorise (Comprehension)*

Six summary figures, sorted into: Answers the complaint / Sounds relevant but doesn't / Actively misleading.

Item	Correct Bin	Why
Detention-petition rate by language group (~30% both)	Sounds relevant but doesn't	Equal petition rates say nothing about whether the underlyi biased before legal review corrected it.
Average compliance-risk score by language group (25-point gap)	Answers the complaint	A large score gap with no petition-rate gap is close to a textb model caught by human/legal review.
Vendor's "94% predictive accuracy" marketing claim	Actively misleading	No disclosed test population or method — not evidence abou cohort.
Raw list of patient names attached to scores	Actively misleading (for this task)	A privacy and due-process problem, not an analytical one; di population-level question.
Correlation between interpreter-access delay and outcome	Answers the complaint	If interpreter delay — not language group itself — predicts th model may be encoding an access failure the county can fix, bias.
Standard deviation of scores within each language group	Sounds relevant but doesn't	Spread-within-group doesn't tell you whether groups differ f

New Wrinkle

Mr. Balogun's name appears in the raw list. Sorting it correctly into 'Actively misleading (for this task)' requires re look him up specifically before the aggregate analysis is even framed — a live, personalised version of Act I Scene lesson.

Item	Correct Bin	Why
Facilitator Note Learners who played Act I will almost universally want to click on his name immediately. Let them — the click proc Bridge screen again, reinforcing that individual curiosity is not a substitute for population method.		

SCENE 8 — Filtering the Cohort

Interaction Type	Map Pin Drop + Sketch and Recognise
ISP Framework	*Map Pin Drop + Sketch and Recognise (Problem-solving–Creation)*

Before the real map loads, sketch your predicted pattern: where on the county map do you expect detention petitions to cluster? Then the actual map of 40 flagged patients drops in, pinned against clinic locations and evening-transit coverage.

Task

- Compare your sketch to the real distribution, then connect the petition-rate pattern to either language group directly, or to evening-DOT-slot availability as the mediating factor.

New Sub-Layer — Where Does Case #7 Sit?

An inset shows Mr. Balogun's case on both the map and the score chart: his score (88%) is well above his group' pin sits on a transit desert with no evening clinic hours — wherever you placed his final remedy in Act I. If the rem with documented rationale,' his case supports the 'legal review corrects the model's structural blind spot' readi 'petition for detention,' his case is an outlier within the very pattern you are about to identify — worth flagging, r

Feedback

Most learners' sketches cluster petitions on language group alone; the real map clusters them on evening-trans correlates with but is not identical to language group. The gap between your sketch and the data is the lesson.

Facilitator Note

Prediction-before-reveal converts a passive chart-read into a genuine test of the learner's working hypothesis. unlocks only if the strongest reading (transit/interpreter access) is selected: 'If legal review is correcting the m now, what happens the day interpreter staffing is cut?' Naming this continuity risk banks a point for Scene 10.

SCENE 9 — The Advocate Calls

Interaction Type	Dialogue Choice / Role-Play
ISP Framework	*Dialogue Choice / Role-Play (Creation)*

Three openings: honest-partial / stonewall / false-reassurance. Partway through, if you chose the honest-partial opening, the advocate says: “One of the eleven is a patient named Tunde — his co-worker mentioned the analyst who came out to see him took the time to actually get an interpreter, whatever else comes of this. Was that you?”

Three Responses

- Confirm, redirect to the aggregate finding ("That was one case among many I'm now looking at in aggregate — here's what the eleven together show...") — keeps the register population-level without denying the personal thread.
- Lean into the personal connection ("Yes — and honestly his case is a good example of the pattern...") — conflates a single case with population evidence, exactly what Scene 7's raw-list item warned against.
- Deflect entirely ("I can't discuss individual patients.") — legally cautious, reads as cold to an advocate making a human gesture, not a legal request.

Decision Consequence

- The strongest move keeps the register matched to whichever question is actually being asked — legal confidentiality where required, honest acknowledgement where possible — and redirects gracefully rather than merging or refusing the personal thread.

SCENE 10 — Competing Deadlines

Interaction Type	Resource Allocation
ISP Framework	*Resource Allocation (Problem-solving)*

72 hours distributed across: deeper cohort analysis, individual file spot-checks, drafting the board report, and meeting with interpreter services about access delays. No zero-cost option.

New Sub-Note

Spot-check hours can pull Mr. Balogun's file. A well-annotated Scene 2, an attached Scene 3 social-worker note Scene 4 record produce a spot-check that confirms the aggregate pattern with a strong, citable example. A mir file produces a spot-check that is merely consistent with the pattern but adds no evidentiary weight — the sanc difference explicitly.

SCENE 11 — Building the Decision Rule

Interaction Type	Conditional Simulator + Stack and Build (Jigsaw) + Rotate to Align
ISP Framework	*Conditional Simulator + Jigsaw Assembly + Rotate to Align (Creation)*

Assemble a policy from three interlocking pieces, checked against the evidence file:

- Legal threshold: keep current petition criteria / raise the evidentiary bar for petitions where interpreter access was unresolved at flag time
- Safeguard: mandatory interpreter-access and evening-DOT-availability check before any petition is filed
- Communication plan: internal-only memo / public transparency report to the coalition and the board

Rotate to Align: the three pieces must be turned into agreement with each other — a safeguard that contradicts the chosen legal threshold, or a communication plan that promises more than the threshold delivers, will not lock into place.

New Check — Exhibit 7 Consistency

The sandbox compares your Act II recommendation against your own Act I remedy for Mr. Balogun. If Act I ended documented rationale' and Act II recommends the raised evidentiary bar plus transparency reporting: "Consiste remedy and your cohort recommendation tell the same story." If Act I ended in 'petition for detention' but Act II's shows legal review is generally correcting the structural bias: "Case #7 was treated more urgently than your ow suggests the pattern warrants. This may be worth revisiting — the sandbox will not force you to, but a board tha might ask why."

Facilitator Note

This is where the design pays off structurally — the learner confronts whether their own two judgements, made pressures and time horizons, actually agree with each other.

SCENE 12 — The Oversight Board Meeting

Interaction Type	Timeline Placement
ISP Framework	*Timeline Placement (Comprehension–Reflection)*

Five outcome cards placed on a four-week timeline, generated from the Scene 11 recommendation and accumulated scores.

New Card, Always Included

"Mr. Balogun's case is referenced by name in the board minutes as [Exhibit 7 / a well-documented example / a ca ask follow-up questions about], depending on your Act I documentation choices."

Facilitator Note
If a workshop only has time to debrief one screen, this is the one.

Bonus Branching Scenarios (Optional Extensions)

Bonus Scene A — The Vendor's Rebuttal

Unlocks
Only if Legal Rigor is high at the end of Scene 8.

ComplianceWatch's vendor disputes the score-gap finding with a narrower time window and a quietly-excluded denominator. Correctly naming the specific manipulation outscores a generic 'the vendor is biased' response.

If you attached (not summarised) the Scene 3 social-worker note, an extra prompt appears: the vendor's deck cites 'insufficient documentation across flagged cases.' Citing Mr. Balogun's attached note as a direct counterexample is worth an additional point unavailable to learners who summarised.

Bonus Scene B — The Second County

Unlocks
After any full playthrough.

A neighbouring county's dataset shows no meaningful score gap. Using Pinch and Zoom to compare the two datasets side by side, the only sandbox-accepted answer is recommending site-by-site checking rather than generalising either way.

Bonus Scene C — The Board Paragraph, Personalised

Unlocks
Only after completing both acts.

Using Voice or Speech Input, record (or type) the one paragraph of the board report addressing Mr. Balogun's case specifically, knowing it will be read by the board and, eventually, by him. Scored on whether it accurately reflects both the individual legal evidence and the population finding, without overstating his case as representative of all eleven or erasing him as an individual within the aggregate.

Feedback & Coaching Notes

Decision Point	Preferred Response	Core Risk of Alternatives
Triage (Sc.1)	Context-weighted sort + documented rationale	Score-only sorting anchors bias; undocumented to Act II
Reading the score (Sc.2)	Separate reliable vs. unreliable features + annotate	Treating all features as equally damning repeats error
Paper trail (Sc.3)	Connect order-schedule to missed visit, attach the note	Reading the flag in isolation overstates urgency; evidentiary weight later
Home visit (Sc.4)	Process-first opening, nuanced write-up	Order-first framing yields worse evidence; comp due-process nuance
Trade-off weighting (Sc.5)	Balanced, documented reasoning	Any single-value extreme creates a previewable comparable cases
Final decision (Sc.6)	Match remedy to evidence category	Mismatched urgency creates both legal and coh
The Bridge	(no action — observe)	Attempting to revise the record defeats the exer
Cohort triage (Sc.7)	Sort by what answers the complaint, resist the name pull	One visible name is not a shortcut past the aggr
Map / cross-tab reasoning (Sc.8)	"Legal review is catching a structural bias the model introduces"	Either extreme reading ignores one chart or the
Advocate call (Sc.9)	Honest-partial, register-matched	Conflating personal and population threads und
Resource allocation (Sc.10)	Balanced spread	Spot-check value is capped by Act I documentat
Decision rule (Sc.11)	Raised evidentiary bar + transparency, checked for consistency against Act I	An inconsistent, unflagged pair of remedies is a

End-of-Simulation Debrief

Narrative Outcomes

A compound ending built from both acts' running scores. Two combinations are singled out:

- **Rigorous & Fair, Twice — the calibration case for next training cycle.**
- **Lucky, Not Ready — Act II's analysis happened to hold up despite a thin Exhibit 7; named explicitly as luck, not method.**

Structured Debrief Themes

- A compliance score is a starting hypothesis, not a legal finding — at both scales.
- Process evidence (a missed shift, an unfulfilled interpreter request) is usually more diagnostic than the flagged moment alone — and only usable later if preserved, not merely noted.
- How you open a conversation determines what evidence you gather; how you write it down determines what someone else, including a board, can use.
- A population-level disparate-impact pattern and a single-case legal judgement answer different questions — pulling one to answer the other's question is this simulation's core failure mode.
- There is rarely one correct trade-off between safety, liberty, consistency, and speed — the skill is making that trade-off visibly and defensibly.
- Legal and ethical decisions have a half-life measured in scale, not just time.

Reflection Questions

- Which of your Scene 1 sorting instincts would you actually use under real hearing-calendar pressure?
- What single piece of process evidence could you start requesting and preserving — not just noting — before any flag occurs in your own setting?
- Where do you sit on the safety/liberty/consistency/speed balance — and did seeing your Act I choices reappear as cohort evidence shift that?

Practical Takeaways

- Build a ninety-second documentation habit into every first-line legal/compliance review — it is the cheapest insurance against a future disparate-impact finding.
- Treat any predictive score's disclaimer section as load-bearing evidence, not fine print.
- Before generalising a pattern across a population, check for a mediating structural variable (access, language, transport) before accepting a demographic explanation.

Facilitator / Developer Notes

- No MCQs throughout. Every decision is embedded in a spatial, dialogic, or allocation mechanic so the 'obvious answer' cannot be telegraphed.
- Why the Bridge has no interaction: the single most load-bearing design decision in the simulation. Don't add a choice here.
- Why Mr. Balogun's case is reused rather than replaced: a generic Act II dataset would still teach cohort-level reasoning, but would leave the central lesson — that individual documentation becomes population legal evidence — merely told to the learner rather than experienced as a direct consequence.
- Facilitator choice — same analyst or a colleague in Act II: same-person runs emphasise self-accountability (good for self-paced training); different-colleague runs reproduce the real institutional experience of inheriting someone else's file (good for group workshops, tends to generate the most candid discussion).
- Every sandbox module (Map Pin Drop, Tilt and Balance, Rotate to Align, Voice/Speech Input, etc.) writes to and reads from the same underlying case-and-cohort dataset object — build it as one persistent data model, not twelve separate mini-app states, so that Act I choices are genuinely retrievable in Act II rather than hard-scripted to appear.

Scoring / Self-Check Matrix

Four running dimensions across both acts (Evidentiary + Legal Rigor merge into Rigor; Procedural Fairness + Stakeholder Trust merge into Trust).

Scene	Rigor	Trust	Consistency / Credibility	E
Sc.1 — The Flag Arrives	context sort +2, note +1	—	—	fa (c
Sc.2 — Reading the Score	correct bin +3, annotation +1/item	disclaimer included +2	—	—
Sc.3 — The Paper Trail	link connected +3, attached +2	—	—	—
Sc.4 — The Home Visit	process-first +1	process-first +3, nuanced write-up +2	—	—
Sc.5 — Weighing It All	—	balanced +3	weighting >2/10 +3	s >
Sc.6 — The Decision	evidence-matched remedy +3	evidence-supported remedy +2	remedy matches precedent-check +3	—
Sc.7 — First Login	correct sort +2	—	—	—
Sc.8 — Filtering the Cohort	strongest reading +3	—	—	—
Sc.9 — The Advocate Calls	register-matched +1	honest-partial +3, follow-up handled +2	—	—
Sc.10 — Competing Deadlines	balanced allocation +2	interpreter-services hours funded +1	spot-check hours (scaled by Act I docs) +0–3	a
Sc.11 — Building the Rule	consistency check passed +3	direct board response +2	Exhibit 7 check passed +3	—
Sc.12 — The Oversight Board	—	—	epilogue reviewed +2	—

Self-Check Prompt

Identify the one Act I choice you made fastest, with the least conscious deliberation. When you saw it reappear in the Bridge or Act II, did it hold up — or did you wish, watching it become someone else's legal evidence, that you'd spent ninety more seconds on it the first time?